ICT Express 12 (2026) 487–492

Contents lists available at ScienceDirect

ICT Express

journal homepage: www.elsevier.com/locate/icte

Text-guided diffusion-based restoration of extremely compressed backgrounds for VCM

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ARTICLE INFO

Keywords:
Diffusion model
Image restoration
Image generation
Video coding for machines (VCM)

ABSTRACT

Restoring high-quality images from severely degraded inputs is essential for video coding for machines (VCM), where background regions are compressed at extremely low bitrates. In this letter, we propose a novel text-guided diffusion-based restoration (TGDR) algorithm, which integrates semantic information from text captions to guide the restoration process. Specifically, we develop a refinement block that incorporates a transformer-based time-aware feature extractor to fuse visual features, time-step embeddings, and textual semantics adaptively to guide a pretrained diffusion model during the reverse denoising process. By incorporating both visual and textual information, TGDR effectively reconstructs complex structures and improves semantic consistency in highly compressed regions. Experimental results show that TGDR achieves superior performance compared to state-of-the-art algorithms.

1. Introduction

Video compression technologies [1–3] have been developed to represent pixel signals for human perception, optimizing storage and transmission efficiently while maintaining visual quality. With the proliferation of smart-city infrastructures and Internet-of-Things (IoT) devices [4,5], the volume of video data transmitted for machine vision applications is rapidly increasing [6]. However, traditional compression techniques often discard high-frequency details and texture information, which are critical for accurate machine analysis [7]. Therefore, new research efforts have recently been made to compress videos, facilitating subsequent high-level vision tasks, known as video coding for machines (VCM) [8–10].

Because several applications still demand high reconstruction quality for human vision for supervision, VCM has been developed in a scalable coding framework to support both human and machine vision tasks [11,12]. In this framework, a compact representation of the frame for machine vision is transmitted in the base layer, and additional information to improve the perceptual quality for human vision is transmitted in the enhancement layer. For example, in [13–17], regions of interest (ROIs), defined as object boundaries, are encoded with high quality for machine vision and transmitted in the base layer, whereas the background (BG) region for human vision is compressed with an extremely low bitrate and transmitted in the enhancement layer. However, the reconstructed frame suffers from severely degraded BGs in this approach, making it unsuitable for human perception.

The main challenge in this scalable coding approach lies in restoring the highly degraded BG resulting from aggressive compression by exploiting high-quality ROIs [18]. In [15–17], visual signals extracted from the degraded image are exploited; however, the structural loss due to extreme compression causes a loss of semantic consistency between the BG and ROIs. A promising direction is to use external semantic priors to guide the restoration process and compensate for the missing structure. In particular, recent text-conditioned algorithms [19–22] inject semantic priors via text captions, demonstrating that well-integrated textual information can provide global context for reconstructing realistic content, even in severely degraded BGs. Moreover, text is lightweight, easily compressible, and transmittable at a minimal bitrate, making it a practical form of auxiliary information in resource-constrained VCM scenarios. However, these approaches assume spatially uniform input quality and are therefore not directly applicable to VCM, where the input image exhibits significant quality variations between the BG and ROIs.

In this work, we propose the first textual semantic information-guided compressed BG restoration algorithm for VCM, called TGDR (**T**ext- **G**uided **D**iffusion-based **R**estoration). Unlike existing approaches

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https://doi.org/10.1016/j.icte.2026.01.011
Received 14 June 2025; Received in revised form 20 January 2026; Accepted 26 January 2026
Available online 3 February 2026
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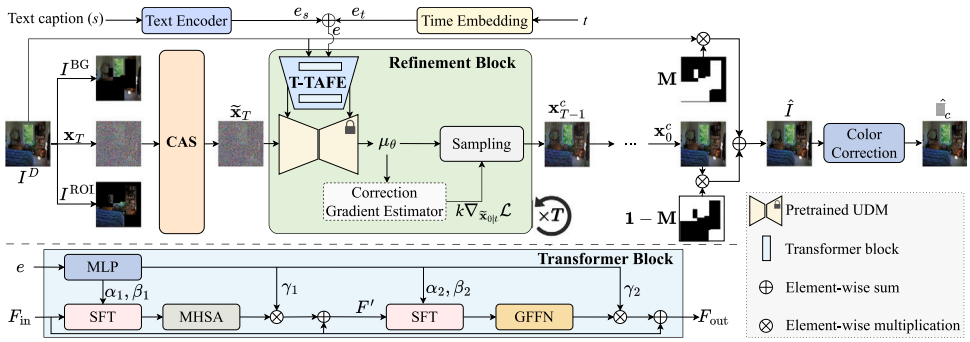


Fig. 1. Overview of the proposed TGDR algorithm. The transformer block in T-TAFE is detailed below the dashed line.

[16,17], TGDR is the first to explicitly incorporate textual semantic information into a structured diffusion framework to address the unique challenges of VCM. Specifically, using a novel refinement block, TGDR controls the output of a pretrained unconditional diffusion model (UDM) at each denoising step. A key component of the refinement block is a transformer-based time-aware feature extractor (T-TAFE), which adaptively captures multiscale features by combining visual input, time-step information, and semantic guidance. This integration enables the embedding of global text semantics into the visual feature space, guiding the reconstruction of realistic structures in severely degraded BGs. Experimental results demonstrate that TGDR significantly outperforms state-of-the-art algorithms.

In summary, we make the following contributions.

- We propose a text-guided diffusion algorithm that restores a high-quality image from a single decoded input containing high-quality ROIs and extremely degraded BGs.
- We develop a learnable conditioning module, T-TAFE, injecting structural image features and global semantic context to guide semantically coherent image generation.
- We experimentally demonstrate that the proposed TGDR significantly outperforms state-of-the-art algorithms on multiple datasets.

2. Proposed algorithm

2.1. Mathematical formulation

As the TGDR algorithm is based on the denoising diffusion probabilistic model (DDPM) [23], a generative model with a forward process and a reverse process, we briefly review it.

In the forward process, the noisy image $\mathbf{x}_t$ at any time step t can be directly sampled from the input image $\mathbf{x}_0$ as

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, \sqrt{1-\bar{\alpha}_t}\mathbf{I}\right), \tag{1}$$

where $\bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i)$, and $\{\beta_t \in (0,1)\}_{t=1}^{T}$ denotes a variance schedule.

In the reverse process, the clean image $\mathbf{x}_0$ is recovered by gradually denoising the pure noise $\mathbf{x}_T$. The inverse sampling at each step is performed by

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \sigma_t^2\mathbf{I}\right), \tag{2}$$

where the mean $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(\mathbf{x}_t, t)\right)$ and noise $\epsilon_\theta(\mathbf{x}_t, t)$ are predicted using a learnable network parameterized by θ, and $\sigma_t^2\mathbf{I}$ is a fixed covariance [23]. A clean version $\tilde{\mathbf{x}}_{0|t}$ at the intermediate step t can be approximated as

$$\tilde{\mathbf{x}}_{0|t} = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}}. \tag{3}$$

2.2. Network architecture

Fig. 1 shows an overview of the proposed TGDR algorithm, which reconstructs a high-quality image $\hat{I}_c$ given a decoded image I^D containing high-quality ROIs and an extremely degraded BG. First, an initial noisy image $\mathbf{x}_T$ is generated by applying the forward process in (1) to the input I^D. Then, the content-aware supervision (CAS) block [16] extracts meaningful features from the ROI and BG subimages, I^{ROI} and I^{BG}, and produces a refined noisy input $\tilde{\mathbf{x}}_T$, which provides a more informative and stable initialization for the subsequent reverse diffusion process. The reverse diffusion is then performed solely by the refinement block, which operates iteratively over timesteps $t = T, \ldots, 1$. At each timestep, the refinement block restores the BG guided by two components: a T-TAFE module that injects input features and semantic text information into the pretrained UDM, and a correction gradient estimator that enforces consistency with the degraded input by adjusting the sampling mean. Next, the restored image $\hat{I}$ is obtained by combining the restored BG with the high-quality ROIs from I^D as

$$\hat{I} = \mathbf{x}_0^c \otimes (\mathbf{1} - \mathbf{M}) \oplus I^D \otimes \mathbf{M}, \tag{4}$$

where $\mathbf{x}_0^c$ is the output of the last refinement block, $\oplus$ and $\otimes$ denote element-wise addition and multiplication, respectively, and $\mathbf{M}$ represents a hard binary mask indicating the high-quality ROIs. Finally, the color distribution of $\hat{I}$ is aligned with that of I^D, as in [17], to obtain $\hat{I}_c$.

2.3. CAS block

The reverse (generative) process described in Section 2.1 begins with high-variance sampled noise, which may degrade restoration performance due to its failure to account for the spatial characteristics of I^D. To address this, we adopt the CAS block [16] to extract essential features from both the degraded BG I^{BG} and high-quality ROIs I^{ROI}, as illustrated in Fig. 1. These features preserve informative content from both regions. By leveraging them, the CAS block constructs a refined noisy input $\tilde{\mathbf{x}}_T$, which serves as a more informative initialization and enhances restoration performance.

2.4. Refinement block

Given the noisy image $\tilde{\mathbf{x}}_T$, a pretrained UDM can generate realistic images; however, its random nature may result in discrepancies between the generated and target images. To mitigate this issue, the proposed refinement block guides the generation process of the pretrained UDM at each diffusion step. The refinement block comprises a correction gradient estimator, which enforces consistency using the degradation model, and a T-TAFE module that adaptively extracts textual and visual features to produce more accurate and content-preserving results.

2.4.1. Correction gradient estimator

The correction gradient technique was developed to enhance the consistency of the UDM output by minimizing the discrepancies between the degraded versions of the input and output images [24,25]. Specifically, it measures the distance between the two degraded images and refines the output to align better with the input. In this work, to generate a high-quality image consistent with a given degraded image I^D, we compute its distance to the approximated clean version $\widetilde{\mathbf{x}}_{0|t}$ as

$$\mathcal{L}\left(\widetilde{\mathbf{x}}_{0|t}, I^D\right) = \left\|\left(\widetilde{\mathbf{x}}_{0|t} - I^D\right) \otimes \mathbf{M}\right\|_2^2 + \left\|\left(\mathcal{D}(\widetilde{\mathbf{x}}_{0|t}) - I^D\right) \otimes (\mathbf{1} - \mathbf{M})\right\|_2^2, \tag{5}$$

where $\mathcal{D}(\cdot)$ denotes a degradation model, *i.e.*, compression using VVC [2]. Then, the estimator computes its gradient $\nabla_{\widetilde{\mathbf{x}}_{0|t}} \mathcal{L}\left(\widetilde{\mathbf{x}}_{0|t}, I^D\right)$,[1] which shifts the distribution of the output of the UDM without retraining the diffusion network. Finally, the conditional image $\mathbf{x}_{t-1}^c$ is sampled from a Gaussian distribution with a mean adjusted by the correction gradient [16,17] as

$$\mathbf{x}_{t-1}^c \sim \mathcal{N}\left(\boldsymbol{\mu}_\theta - k\nabla_{\widetilde{\mathbf{x}}_{0|t}} \mathcal{L}\left(\widetilde{\mathbf{x}}_{0|t}, I^D\right), \sigma_t^2 \mathbf{I}\right), \tag{6}$$

where k is a parameter that controls the impact of the gradient. When $k = 0$, $\mathbf{x}_{t-1}^c$ is sampled purely from the distribution of the UDM without guidance from the correction gradient. As k increases, stronger guidance is applied, enforcing a closer alignment with the degraded input.

2.4.2. T-TAFE module

Although the correction gradient $\nabla_{\widetilde{\mathbf{x}}_{0|t}} \mathcal{L}$ in (6) conditions the restoration on the degraded image, directly shifting the mean of the unconditional distribution $\boldsymbol{\mu}_\theta$ may cause instability due to a mismatch between the manifolds of $\boldsymbol{\mu}_\theta$ and $\nabla_{\widetilde{\mathbf{x}}_{0|t}} \mathcal{L}$. Moreover, if the BG of the input I^D is heavily degraded, relying on the correction gradient alone may fail to reconstruct structural details accurately. To address this, we enhance the reverse diffusion by developing a learnable conditioning module, T-TAFE, comprising six transformer blocks. Specifically, the T-TAFE module injects structural image features, time-step embeddings, and a global semantic context into the pretrained UDM using spatially-adaptive denormalization (SPADE) blocks [26], following a strategy similar to that in [16].

In [16,17], a U-Net encoder was employed to condition the reverse process using visual features and time-step information. However, these approaches face two critical limitations: (i) they do not use textual inputs, which contain global semantic information, and (ii) their residual blocks are limited in capturing long-range dependencies, which are critical for restoring semantically coherent content. To overcome these limitations, the T-TAFE module employs transformers and incorporates text embeddings into the denoising process.

As shown in Fig. 1, T-TAFE conditions each reverse diffusion step on a composite embedding e, defined as the sum of the semantic embedding e_s from a pretrained text encoder [27] and the time-step embedding e_t, *i.e.*, $e = e_s \oplus e_t$, which is recomputed at each time step t to ensure temporal alignment. This design eliminates additional projection layers or parameters, improving efficiency. For efficient and spatially-adaptive prior injection, we adopt a two-stage affine modulation strategy [28–30]. Specifically, the composite embedding e is fed into a multi-layer perceptron (MLP) in each transformer block to generate modulation parameters $\{\alpha_i, \beta_i, \gamma_i\}_{i=1}^2$. In the first stage, F_{in} is modulated via a spatial feature transform (SFT) [28] with (α_1, β_1), followed by a multi-head self-attention (MHSA) [31] scaled by γ_1 as

$$F' = F_{\text{in}} + \gamma_1 \otimes \text{MHSA}(\text{SFT}(F_{\text{in}} \mid \alpha_1, \beta_1)). \tag{7}$$

Its implementation details are provided in the supplemental material. In the second stage, F' is further modulated by another SFT using (α_2, β_2), followed by a gating feedforward network (GFFN) [32] scaled by γ_2 to produce output F_{out} as

$$F_{\text{out}} = F_{\text{in}} + \gamma_2 \otimes \text{GFFN}(\text{SFT}(F' \mid \alpha_2, \beta_2)). \tag{8}$$

The complete TGDR algorithm is summarized in Algorithm 1.

Algorithm 1 TGDR algorithm

Input: I^D, k, s, and $\mathcal{D}(\cdot)$.
Sample $\mathbf{x}_T \sim q(\mathbf{x}_T | I^D)$ using (1).
$\widetilde{\mathbf{x}}_T = \text{CAS}(\mathbf{x}_T)$.
for t from T to 1 **do**
  Compute $\widetilde{\mathbf{x}}_{0|t}$ using (3).
  Correction Gradient Estimator
  Compute $\mathcal{L}\left(\widetilde{\mathbf{x}}_{0|t}, I^D\right)$ using (5).
  Compute $\nabla_{\widetilde{\mathbf{x}}_{0|t}} \mathcal{L}\left(\widetilde{\mathbf{x}}_{0|t}, I^D\right)$.
  Sampling
  Sample $\mathbf{x}_{t-1}^c$ using (6).
end for
Compute $\hat{I}$ using (4).
Compute $\hat{I}_c$ [17].
Output: $\hat{I}_c$

2.5. Loss functions

To train the proposed TGDR, we define the loss $\mathcal{L}_{\text{TS}}$ as the weighted sum of the diffusion loss $\mathcal{L}_d$, pixel loss $\mathcal{L}_p$, and semantic loss $\mathcal{L}_s$ as

$$\mathcal{L}_{\text{TS}} = \mathcal{L}_d + \lambda_p \mathcal{L}_p + \lambda_s \mathcal{L}_s, \tag{9}$$

where λ_p and λ_s are hyperparameters that control the balance among the three losses.

We employ the diffusion loss $\mathcal{L}_d$, used for training the diffusion model [23], defined as

$$\mathcal{L}_d = \mathbb{E}_{\mathbf{x}_t, \epsilon, t}\left[\|\epsilon - \epsilon_\theta\|_2^2\right], \tag{10}$$

where ϵ and ϵ_θ denote the true and predicted noise vectors, respectively. The pixel loss $\mathcal{L}_p$ quantifies the l_2-norm of the difference between the ground-truth $\mathbf{x}^{GT}$ and estimated clean image $\widetilde{\mathbf{x}}_{0|t}$ as

$$\mathcal{L}_p = \left\|\widetilde{\mathbf{x}}_{0|t} - \mathbf{x}^{GT}\right\|_2^2. \tag{11}$$

As $\mathcal{L}_d$ and $\mathcal{L}_p$ focus only on low-level accuracy, *i.e.*, pixel values and noise, they lack semantic guidance and often fail to ensure structural coherence under severe degradation. To address this failure and preserve the global semantic information, we define a semantic loss $\mathcal{L}_s$ that measures the distance between visual and textual embeddings as

$$\mathcal{L}_s = \left\|E_I(\widetilde{\mathbf{x}}_{0|t}) - E_s(s)\right\|_2^2, \tag{12}$$

where E_I and E_s denote the image and text encoders, respectively, from a pretrained CLIP model [27]. It is well known that, because CLIP is trained on a large and diverse image–text dataset, its embeddings are well aligned and generalize effectively to new domains [33,34].

3. Experimental results

3.1. Implementation details

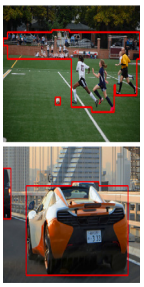
We use a single DDPM, pretrained on ImageNet [25], as the pretrained UDM. During training, only the T-TAFE module and SPADE blocks are optimized, whereas the parameters of the pretrained UDM remain frozen and are used for the learnable conditioning components without any fine-tuning. We use the Adam optimizer with a learning rate of 2×10^{-5} and a batch size of 2 for 100 epochs, which takes approximately three weeks to train on two Nvidia RTX 3090 GPUs. The

[1] Its detailed derivation is provided in the supplemental material.

Table 1
Quantitative comparison on the COCO [35] and OpenImages [36] datasets. For each metric, the best result is **boldfaced**, while the second-best is underlined.

	COCO						OpenImages						Complexity	
	LPIPS↓	FID↓	MUSIQ↑	ARNIQA↑	PSNR↑	SSIM↑	LPIPS↓	FID↓	MUSIQ↑	ARNIQA↑	PSNR↑	SSIM↑	Runtime	# Param. (M)
BrushNet [19]	0.1512	4.37	65.57	0.7169	25.85	0.7366	0.1281	13.94	64.21	0.7246	25.97	0.7751	0.446	1429.5
ControlNet [20]	0.2137	5.94	**66.29**	0.7192	23.29	0.6962	0.2008	19.55	**64.53**	0.7250	22.87	0.7376	0.233	367.2
T2IAdapter [21]	0.2111	6.58	65.69	0.7101	21.55	0.6931	0.2043	22.22	64.18	0.7198	20.33	0.7240	0.366	77.4
PowerPaint [22]	0.1950	6.68	63.62	0.7185	23.58	0.6876	0.1515	17.25	64.27	0.7248	25.10	0.7521	0.287	123.1
GGDR [17]	0.1338	4.30	63.93	0.7262	**27.03**	**0.7893**	0.0748	8.11	63.66	0.7271	30.65	**0.8667**	1.922	670.0
Proposed	**0.1293**	**4.04**	64.53	**0.7312**	26.96	0.7876	**0.0746**	**7.89**	63.82	**0.7285**	**30.66**	0.8653	2.114	673.3

The statistical analysis is provided in the supplemental material.



"Players and a referee are in motion on a sports field, with spectators and vehicles behind them."

"A sports car on a bridge contrasts with city buildings, highlighting urban elegance and modern design."

Fig. 2. Examples of images and their corresponding generated text captions from the (top) COCO [35] and (bottom) OpenImages [36] datasets.

hyperparameters λ_p and λ_s in (9) are set to 1 and 0.01, respectively, based on preliminary experiments to balance stable convergence and high restoration quality. The gradient scaling parameter k in (6) is set to 0.09. The degradation model $D(\cdot)$ in (5) is implemented using VVC [2] with a quantization parameter (QP) value of QP = 52. During inference, we use 50 sampling steps for the reverse diffusion process. The source code and pretrained models will be available on our project website.[2]

3.2. Datasets

The TGDR was trained using only the COCO dataset [35], which contains 70,781 images spanning diverse scenes and includes bounding box annotations that define the ROIs necessary for modeling ROI-BG relationships in restoration. For evaluation, we use 4990 images from the official COCO validation set, as it provides consistent ROI bounding box annotations. To further assess generalization to unseen scenes with different data distributions, we additionally evaluate on 1000 randomly selected images from the OpenImages dataset [36]. The ground-truth bounding boxes in both datasets are used to define the ROIs, *i.e.*, $\mathbf{M}$; however, any object detection algorithm can also be used in practice. All input images are resized to 256 × 256 pixels using bicubic interpolation, and the restored images are produced at the same resolution. This preprocessing step is uniformly applied across all algorithms for a fair comparison.

We generate text captions using GPT-4o by providing images with ROIs and the following query: "Describe the background of this image to highlight the semantic relationship between the ROIs (red boxes) and the background in a single sentence (maximum 20 words)". Fig. 2 shows randomly selected examples from the COCO and OpenImages datasets and their corresponding generated captions. The evaluation of their semantic consistency and its impact on performance is provided in the supplemental material.

3.3. Comparison with state-of-the-art algorithms

We evaluate the performance of the proposed algorithm against several recent state-of-the-art algorithms. BrushNet [19], ControlNet [20], T2IAdapter [21], and PowerPaint [22] represent the latest generation of text-conditioned diffusion models and are serve as strong baselines for text-guided image restoration. We also include GGDR [17], a diffusion-based restoration framework designed for extreme compression, as the strongest non-text baseline aligned with the VCM dataset. For a fair comparison, all algorithms were retrained with the recommended settings using our dataset, containing high-quality ROIs and low-quality BGs.

Table 1 compares the quantitative performance using six quality metrics: LPIPS, FID, MUSIQ [37], ARNIQA [38], PSNR, and SSIM. LPIPS measures perceptual similarity via deep feature distances, while FID evaluates distributional realism by comparing feature statistics between restored and ground-truth sets. MUSIQ provides a transformer-based no-reference perceptual quality score, and ARNIQA is a distortion-aware no-reference image quality metric. PSNR and SSIM assess pixel-level fidelity and structural consistency, respectively. Lower LPIPS and FID scores imply better perceptual quality, whereas higher MUSIQ, ARNIQA, PSNR, and SSIM values indicate better reconstruction quality. On both datasets, the proposed TGDR outperforms all text-conditioned algorithms in all metrics except for MUSIQ, where ControlNet achieves the best scores. However, as will be discussed later, ControlNet has poor alignment and structural accuracy. In contrast, the proposed TGDR achieves superior LPIPS, FID, and ARNIQA scores and competitive PSNR and SSIM values, demonstrating balanced performance in perceptual quality and structural consistency. Furthermore, the proposed TGDR achieves overall the best performance on the OpenImages dataset, which is not seen during training, demonstrating its strong generalization ability. Table 1 also compares the computational complexity in terms of the average runtime and the number of network parameters of different algorithms. The results demonstrate that the proposed TGDR achieves a superior performance-complexity trade-off to conventional algorithms.

Fig. 3 compares the restoration results obtained by each algorithm for a test image in the COCO dataset.[3] The conventional text-conditioned algorithms in Figs. 3(c)–(f) generate content approximately aligning the text but exhibit structural inaccuracy and visible artifacts, *e.g.*, the storage shelf and carpet. In Fig. 3(g), GGDR better preserves structural details, despite lacking textual guidance, due to its degradation-aware prior; however, it exhibits limited semantic fidelity, *e.g.*, incorrect objects on the shelf. In contrast, the proposed TGDR in Fig. 3(h) achieves superior semantic alignment and visual coherence, reconstructing realistic textures and consistent geometry. This demonstrates its effectiveness in leveraging the input image and textual guidance via T-TAFE to restore severely degraded BGs.

[2] https://github.com/hueledao/TGDR.

[3] More qualitative comparisons are provided on our project website.

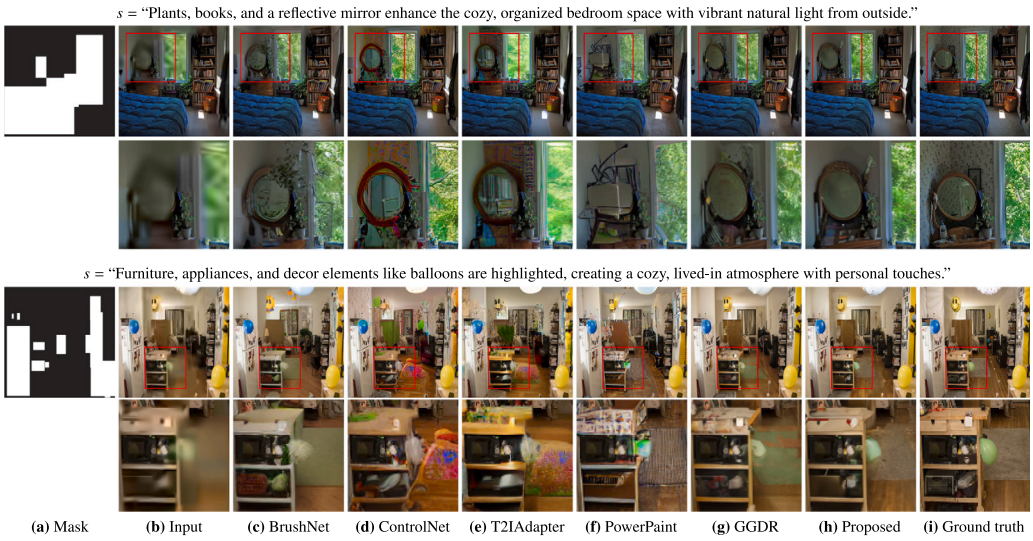


Fig. 3. Qualitative comparison of restored images on the COCO dataset. Magnified regions in the red boxes are shown for detail. The input image in (b) with the mask in (a) is restored using each algorithm in (c)–(h).

Table 2
Effects of T-TAFE, e_s, and $\mathcal{L}_s$ on performance.

T-TAFE	e_s	$\mathcal{L}_s$	LPIPS	FID	MUSIQ	ARNIQA	PSNR	SSIM
			0.1338	4.30	63.93	0.7262	**27.03**	**0.7893**
✓			0.1330	4.11	64.44	0.7308	26.92	0.7891
✓	✓		0.1332	4.04	64.15	0.7286	26.72	0.7859
✓	✓	✓	**0.1293**	**4.04**	**64.53**	**0.7312**	26.96	0.7876

The statistical analysis is provided in the supplemental material.

(a) Mask (b) Input (c) Output (d) GT

Fig. 4. Example of a failure case of the proposed algorithm.

3.4. Component analysis

We analyze the effectiveness of T-TAFE, text embedding e_s, and semantic loss $\mathcal{L}_s$ on performance by training the proposed algorithm with different settings. Table 2 compares the results. The T-TAFE module improves performance by extracting features effectively from both visual and semantic information while capturing long-range dependencies from the degraded input. Incorporating the text embedding e_s further improves performance by providing complementary guidance in degraded regions, enriching contextual details. In addition, the semantic loss $\mathcal{L}_s$ contributes to performance gains by preserving the global semantic information. Finally, combining all components achieves the best performance, indicating their complementary effectiveness.

3.5. Limitations

Although the proposed TGDR effectively restores degraded images in most cases, it depends on accurate ROI detection. Specifically, if salient objects are mistakenly classified as BG and heavily compressed, TGDR may restore them inaccurately. Fig. 4 shows such an example, where the region highlighted by the red rectangle in Fig. 4(b) indicates an erroneous detected ROI and is thus restored inaccurately in Fig. 4(c). However, since this failure arises from erroneous ROI detection, employing a more accurate ROI detection algorithm could improve the restoration performance.

4. Conclusion

We proposed the text-guided diffusion-based algorithm, TGDR, to restore extremely compressed BGs for VCM. Integrating semantic information from text captions with visual information via the proposed T-TAFE module allows TGDR to have fine-grained control of a pre-trained diffusion model at each denoising step. This integration enables the restoration of realistic textures and maintains structural consistency in degraded regions. Experimental results confirmed that the proposed TGDR outperforms state-of-the-art algorithms, demonstrating its effectiveness for VCM applications.

CRediT authorship contribution statement

Le Thi Hue Dao: Writing – original draft, Software, Methodology, Investigation, Data curation. **Naeun Yang:** Writing – review & editing, Data curation. **Jooyoung Lee:** Writing – review & editing, Project administration, Conceptualization. **Seyoon Jeong:** Project administration, Conceptualization. **Chul Lee:** Writing – review & editing, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

Acknowledgments

This work was supported in part by Institute of Information & Communications Technology Planning & Evaluation (IITP), South Korea grant funded by the Korea government (MSIT) (No. 2020-0-00011, Video Coding for Machine) and in part by the Dongguk University Research Fund of 2025.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.icte.2026.01.011.

References

[1] G.J. Sullivan, J.-R. Ohm, W.-J. Han, T. Wiegand, Overview of the high efficiency video coding (HEVC) standard, IEEE Trans. Circuits Syst. Video Technol. 22 (12) (2012) 1649–1668.

[2] B. Bross, Y. Wang, Y. Ye, S. Liu, J. Chen, G.J. Sullivan, J.-R. Ohm, Overview of the versatile video coding (VVC) standard and its applications, IEEE Trans. Circuits Syst. Video Technol. 31 (10) (2021) 3736–3764.

[3] A. Dhanalakshmi, L. Balaji, C. Raja, J. Giri, M. Alrashoud, Residue super-resolution convolutional neural network based complexity reduction for H.266 VVC intra-coding, ICT Express 11 (3) (2025) 460–466.

[4] M.N. Ramadan, M.A. Ali, S.Y. Khoo, M. Alkhedher, Federated learning and TinyML on IoT edge devices: Challenges, advances, and future directions, ICT Express 11 (4) (2025) 754–768.

[5] J. Park, J. Baek, Y. Song, Optimizing smart city planning: A deep reinforcement learning framework, ICT Express 11 (1) (2025) 129–134.

[6] I. Yaqoob, M.U. Hassan, D. Niu, X. Zhao, I.A. Hameed, S.-U. Hassan, A novel person re-identification network to address low-resolution problem in smart city context, ICT Express 9 (5) (2023) 809–814.

[7] H. Wang, X. Wu, Z. Huang, E.P. Xing, High-frequency component helps explain the generalization of convolutional neural networks, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2020, pp. 8684–8694.

[8] L. Duan, J. Liu, W. Yang, T. Huang, W. Gao, Video coding for machines: A paradigm of collaborative compression and intelligent analytics, IEEE Trans. Image Process. 29 (2020) 8680–8695.

[9] M. Rafie, L. Yu, Y. Zhang, S. Liu, Call for Evidence for Video Coding for Machines, output document, m55605, ISO/IEC JTC 1/SC 29/WG 2, 2020.

[10] W. Yang, H. Huang, Y. Hu, L.-Y. Duan, J. Liu, Video coding for machines: Compact visual representation compression for intelligent collaborative analytics, IEEE Trans. Pattern Anal. Mach. Intell. 46 (7) (2024) 5174–5191.

[11] S. Yang, Y. Hu, W. Yang, L.-Y. Duan, J. Liu, Towards coding for human and machine vision: Scalable face image coding, IEEE Trans. Multimedia 23 (2021) 2957–2971.

[12] H. Choi, I.V. Bajić, Scalable image coding for humans and machines, IEEE Trans. Image Process. 31 (2022) 2739–2754.

[13] C. Hollmann, S. Liu, M. Rafie, Y. Zhang, Use Cases and Requirements for Video Coding for Machines, output document, n161, ISO/IEC JTC 1/SC 29/WG 2, 2022.

[14] O. Stankiewicz, S. Różek, M. Lorkiewicz, T. Grajek, S. Maćkowiak, M. Wawrzyniak, J. Stankowski, M. Domański, Background Transmission using RoI-Based Retargeting, output document, m70542, ISO/IEC JTC 1/SC 29/WG 4, 2024.

[15] L.T.H. Dao, A.G. Vien, J. Lee, S. Jeong, C. Lee, Restoration of extremely compressed background for VCM using guided generative priors, in: Proc. IEEE Int. Conf. Image Process., 2023, pp. 1190–1194.

[16] L.T.H. Dao, A.G. Vien, J. Lee, S. Jeong, N. Yang, C. Lee, Content-aware supervision for diffusion-based restoration of extremely compressed background for VCM, in: Proc. IEEE Int. Conf. Image Process., 2024, pp. 1683–1689.

[17] L.T.H. Dao, A.G. Vien, J. Lee, S. Jeong, N. Yang, C. Lee, Gradient-guided diffusion-based restoration of extremely compressed backgrounds for video coding for machines, IEEE Trans. Circuits Syst. Video Technol. 36 (1) (2026) 685–701.

[18] S. Ma, X. Zhang, C. Jia, Z. Zhao, S. Wang, S. Wang, Image and video compression with neural networks: A review, IEEE Trans. Circuits Syst. Video Technol. 30 (6) (2019) 1683–1698.

[19] X. Ju, X. Liu, X. Wang, Y. Bian, Y. Shan, Q. Xu, BrushNet: A plug-and-play image inpainting model with decomposed dual-branch diffusion, in: Proc. European Conf. Comput. Vis., 2024, pp. 150–168.

[20] L. Zhang, A. Rao, M. Agrawala, Adding conditional control to text-to-image diffusion models, in: Proc. IEEE/CVF Int. Conf. Comput. Vis., 2023, pp. 3836–3847.

[21] C. Mou, X. Wang, L. Xie, Y. Wu, J. Zhang, Z. Qi, Y. Shan, T2I-Adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models, in: Proc. AAAI Conf. Artif. Intell., Vol. 38, 2024, pp. 4296–4304.

[22] J. Zhuang, Y. Zeng, W. Liu, C. Yuan, K. Chen, A task is worth one word: Learning with task prompts for high-quality versatile image inpainting, in: Proc. European Conf. Comput. Vis., 2024, pp. 195–211.

[23] J. Ho, A. Jain, P. Abbeel, Denoising diffusion probabilistic models, in: Proc. Adv. Neural Inf. Process. Syst., 2020, pp. 6840–6851.

[24] B. Fei, Z. Lyu, L. Pan, J. Zhang, W. Yang, T. Luo, B. Zhang, B. Dai, Generative diffusion prior for unified image restoration and enhancement, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2023, pp. 9935–9946.

[25] P. Dhariwal, A. Nichol, Diffusion models beat GANs on image synthesis, in: Proc. Adv. Neural Inf. Process. Syst., 2021, pp. 8780–8794.

[26] T. Park, M.-Y. Liu, T.-C. Wang, J.-Y. Zhu, Semantic image synthesis with spatially-adaptive normalization, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2019, pp. 2337–2346.

[27] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: Proc. Int. Conf. Mach. Learn., 2021, pp. 8748–8763.

[28] X. Wang, K. Yu, C. Dong, C.C. Loy, Recovering realistic texture in image super-resolution by deep spatial feature transform, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2018, pp. 606–615.

[29] C. Ham, J. Hays, J. Lu, K.K. Singh, Z. Zhang, T. Hinz, Modulating pretrained diffusion models for multimodal image synthesis, in: Proc. ACM SIGGRAPH, 2023, pp. 1–11.

[30] S. Ye, H. Wang, M. Tan, F. Liu, Recurrent affine transformation for text-to-image synthesis, IEEE Trans. Multimedia 26 (2023) 462–473.

[31] H. Tan, X. Liu, B. Yin, X. Li, MHSA-Net: Multihead self-attention network for occluded person re-identification, IEEE Trans. Neural Netw. Learn. Syst. 34 (11) (2022) 8210–8224.

[32] X. Nie, X. Chen, H. Jin, Z. Zhu, Y. Yan, D. Qi, Triplet attention transformer for spatiotemporal predictive learning, in: Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis., 2024, pp. 7036–7045.

[33] A. Fang, G. Ilharco, M. Wortsman, Y. Wan, V. Shankar, A. Dave, L. Schmidt, Data determines distributional robustness in contrastive language image pre-training CLIP, in: Proc. Int. Conf. Mach. Learn., 2022, pp. 6216–6234.

[34] W. Tu, W. Deng, T. Gedeon, Toward a holistic evaluation of robustness in CLIP models, IEEE Trans. Pattern Anal. Mach. Intell. 47 (9) (2025) 8280–8296.

[35] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C.L. Zitnick, Microsoft COCO: Common objects in context, in: Proc. European Conf. Comput. Vis., 2014, pp. 740–755.

[36] I. Krasin, T. Duerig, N. Alldrin, V. Ferrari, S. Abu-El-Haija, A. Kuznetsova, H. Rom, J. Uijlings, S. Popov, A. Veit, et al., OpenImages: A public dataset for large-scale multi-label and multi-class image classification, Dataset 2 (3) (2017) 18.

[37] J. Ke, Q. Wang, Y. Wang, P. Milanfar, F. Yang, MUSIQ: Multi-scale image quality transformer, in: Proc. IEEE/CVF Int. Conf. Comput. Vis., 2021, pp. 5148–5157.

[38] L. Agnolucci, L. Galteri, M. Bertini, A. Del Bimbo, ARNIQA: Learning distortion manifold for image quality assessment, in: Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis., 2024, pp. 189–198.